



De-Havilland DHC-8-400 B1 & B2 TRAINING MANUAL

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CHAPTER ATA 74 IGNITION SYSTEM

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Ignition System – General

Introduction

The ignition system supplies the electrical energy to ignite the fuel/air mixture during engine start-up. It also provides in flight re-light capability in the event of a flame out.

General Description

The system function is performed by the ignition exciters, cables and plugs. The ignition system includes the systems that follow:

- Ignition Power Supply
- Distribution
- Ignition Selection

Detailed Description

[Refer Figure - Ignition System Block Diagram](#)

The PW150A ignition system consists of the ignition exciter, the ignition cables and the ignition plugs. The ignition exciter is a dual channel unit contained in a single housing and is supplied with aircraft 28 VDC, through a single electrical connector for both channels. It outputs a high voltage signal to the igniter plugs when selected “ON”. The ignition exciter is connected to the igniter plugs by a pair of braided cables. The cables are mounted with quick release connectors to facilitate removal and installation. The igniter plugs are located in the gas generator case and are the drop in type.

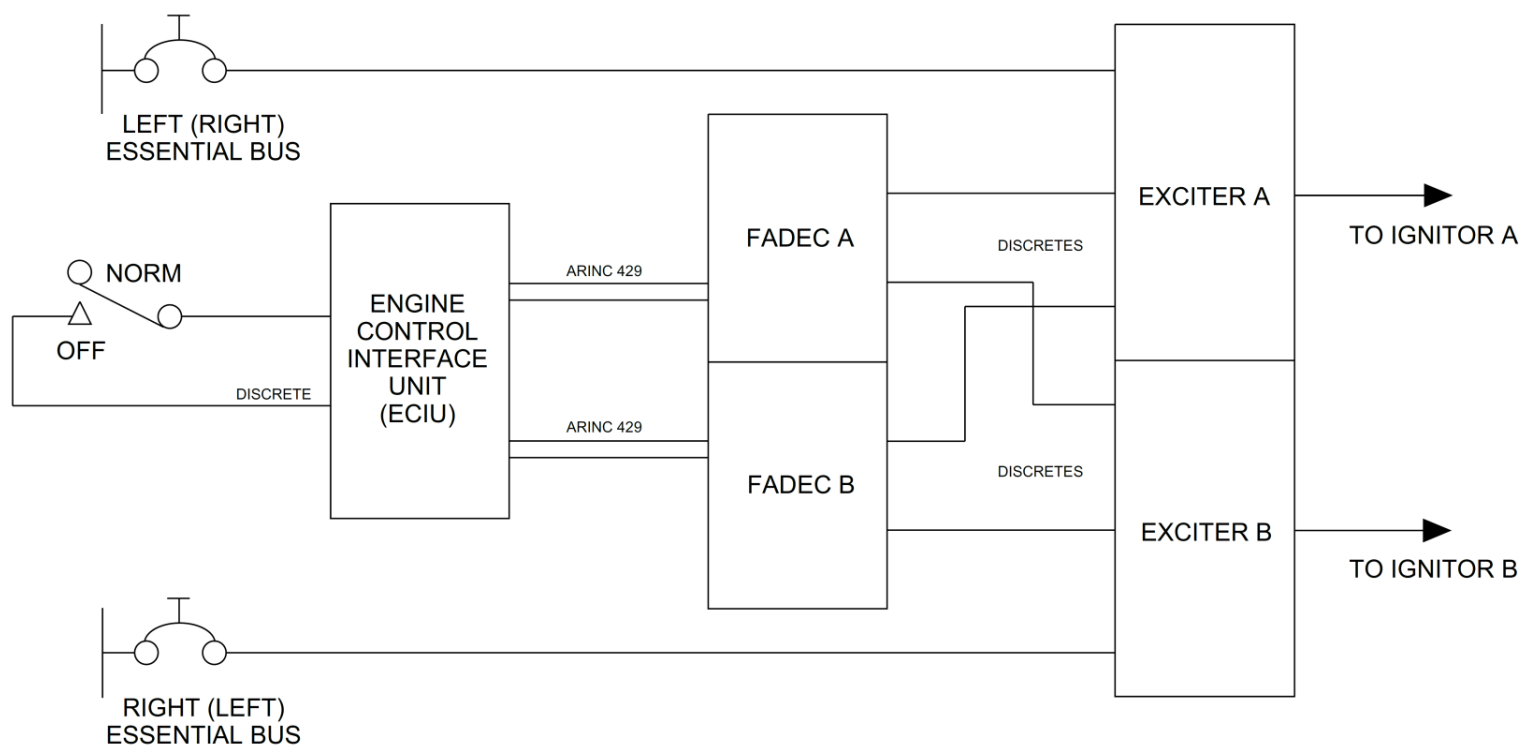


Figure - Ignition System Block Diagram

Ignition Power Supply

Introduction

The ignition exciter transforms a DC voltage input into a pulsed high voltage output to provide the energy to the igniter plugs in order to create the spark that will ignite the fuel/air mixture.

General Description

The function of the ignition power supply is performed by the ignition exciter.

The ignition exciter has the components that follow:

- Ignition Exciter housing
- Ignition Exciter circuitry

Detailed Description

The dual channel ignition system, is powered electrically by the aircraft essential buses and activated by the Full Authority Digital Electronic Control (FADEC). The aircraft essential power buses provide alternate electrical power to the FADEC for engine starting and in the event of a Permanent Magnet Alternator (PMA) malfunction. The PMA has independent coils that provide electrical power to the individual channels of the FADEC when High Pressure Turbine Speed (NH) is above 20% minimum.

The ignition system is configured such that both ignitors can be commanded by each FADEC channel, while maintaining electrical isolation between the FADEC channels and the aircraft essential buses. Only one of the two ignitors is turned on during a normal start (this is to identify any failures in the dual channel ignition system by cycling between ignitors used on subsequent ground starts).

When power is supplied to the ignition exciter the unit's circuitry will raise the energy of this power to a high enough level to cause the ignition plugs to spark. These sparks are used to ignite the fuel/air mixture.

The ignition exciter is a sealed unit and cannot be repaired in the field.

Ignition Functional Test

In order to verify the Ignition system, the FADEC performs an Ignition Test Function when in Maintenance Mode:

With the FADEC in Maintenance Mode, and the Ignition Selection on the Engine Start Panel in the "Normal" position, set the PLA to Flight Idle (FI) detent and the Condition Lever Angle (CLA) in the Shutdown position, selection of the Max Power Button on the Engine Control Panel will cause the FADEC to command the Ignitors to operate.

By disabling the power supply to ignition exciter box (through cockpit circuit breakers) each ignitor can be tested individually.

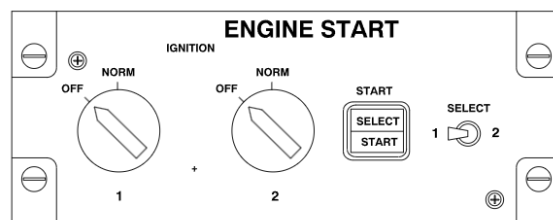
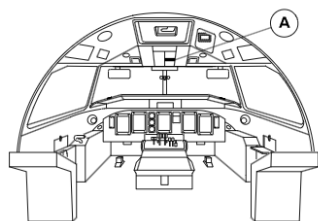
Ignition Audible Check for FADEC S/W D809.3 (POST SB84-73-02 OR POST MODSUM 4-113553)

[Refer Figure - Engine Start Panel](#), [Engine Maintenance Panel](#) & [Engine Control Panel](#)

In order to verify that the engine Ignition system associated wiring is correctly installed, select the ignition "ON" and listen for the "Sparking" noise.

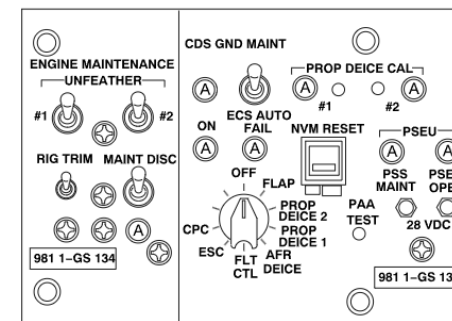
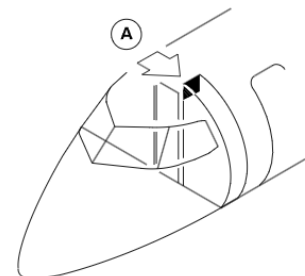
With the Engines "OFF" and FADEC in Maintenance Mode, set the PLA for both engines to "Flight Idle" (FI) position. With the Condition Lever Angle (CLA) in the Shutdown position, set the Ignition to the "Normal" position on the Engine Start Panel for the engine under test. Select "MAINT DISC" button to "ON" position from the Engine Maintenance Panel. The ITT indication should read 492.8 °F (256 °C) confirming that the FADEC is in Maintenance mode. Select "MCR" on the Engine Control Panel.

The Igniters will be commanded to fire sequentially for 10 second intervals. The ITT gauge will indicate 32 °F (0 °C) when Ignitor A is commanded to fire and 3257.6 °F (1792 °C) when Ignitor B is commanded to fire. Check igniters are firing (for the engine under test) by audible check in the vicinity of combustor area. Ignition audible check can be done for the other engine by repeating the procedure.



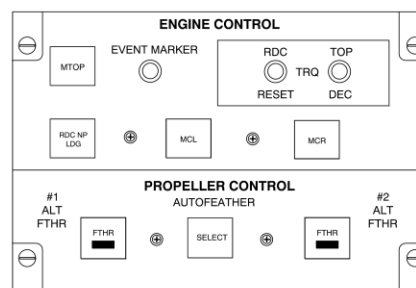
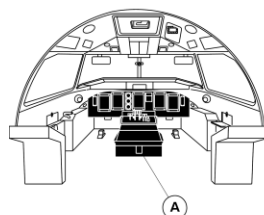
A

Figure 74-2 Engine Start Panel



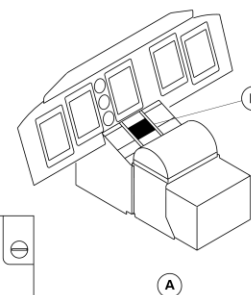
A

Figure 74-3 Engine Maintenance Panel



B

Figure 74-4 Engine Control Panel



Ignition Power Supply – Component Description

Ignition Exciter

[Refer Figure - Ignition Exciter](#)

The Ignition Exciter Box is a dual channel unit contained in a single enclosure. The Ignition Exciter Box is supplied with aircraft 28 Vdc through a single electrical connector, for both channels. It outputs a high-tension signal to the igniter plugs when selected "ON " by the FADEC. The Ignition Exciter Box is connected to the Igniter Plugs by a pair of braided cables. Each FADEC channel (A/B) can fire either ignitors A or B through discrete inputs to the Ignition Exciter Box.

The ignition exciter is flexibly installed on the right hand side of the engine. It is a sealed unit that contains electronic circuitry which provides a high voltage output (15 to 20 kVA) that is transmitted by the ignition cables to the igniter plugs. This high voltage output causes the ignition plugs to spark at a rate of 1 to 1.4 sparks per second.

The two Igniter Plugs are located in the Gas Generator Case and are connected to the Ignition Exciter Box. They provide sparks for igniting the fuel/air mixture.

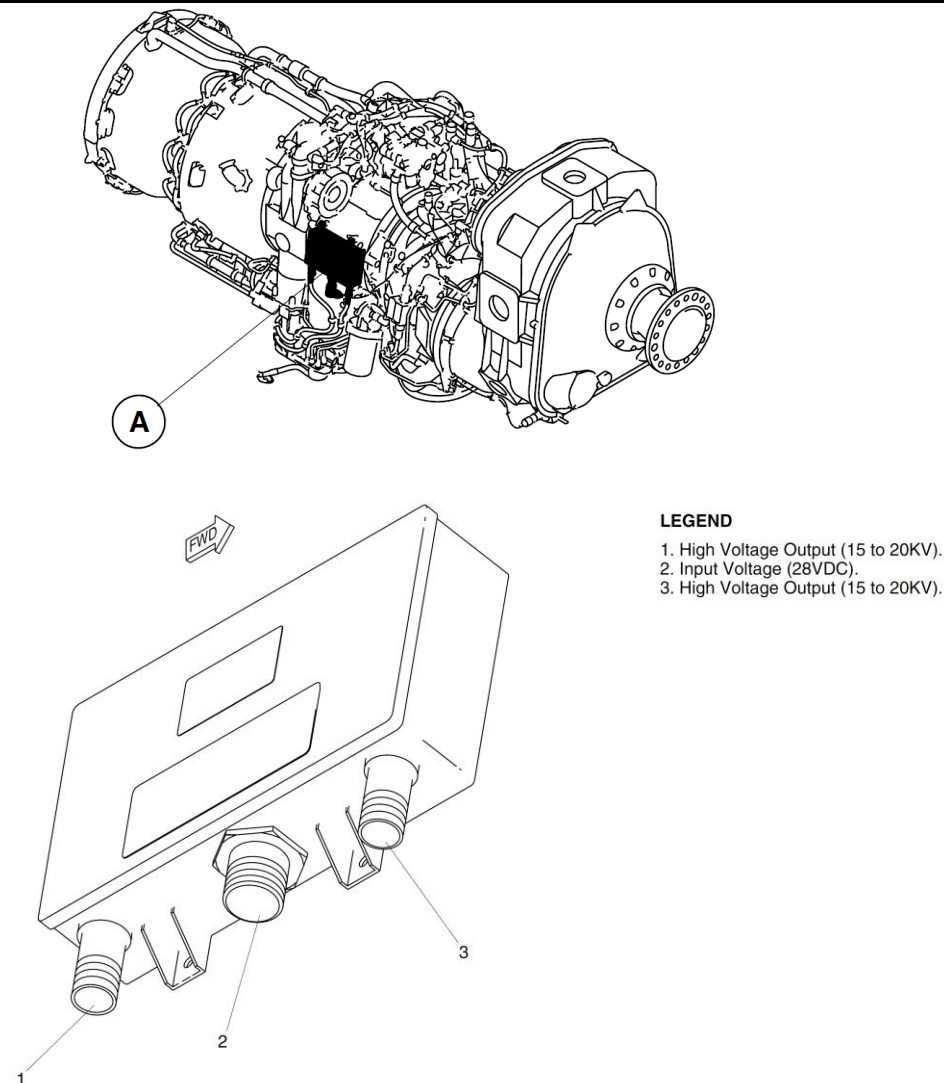


Figure - Ignition Exciter

Ignition Exciter Circuitry and Operation

[Refer Figure - Ignition Power Supply Schematic](#)

When the unit is energized, a capacitor on the high voltage side of the output transformer is progressively charged. When sufficient energy to ionize a spark gap in the unit is accumulated, the capacitor discharges through a dividing

and step-up transformer network across the two ignition plugs. The unit will continue functioning if one of the plugs is open or shorted, in order to operate the remaining plug. If both ignition plugs fail, or if the input voltage is switched off, the capacitor automatically discharges.

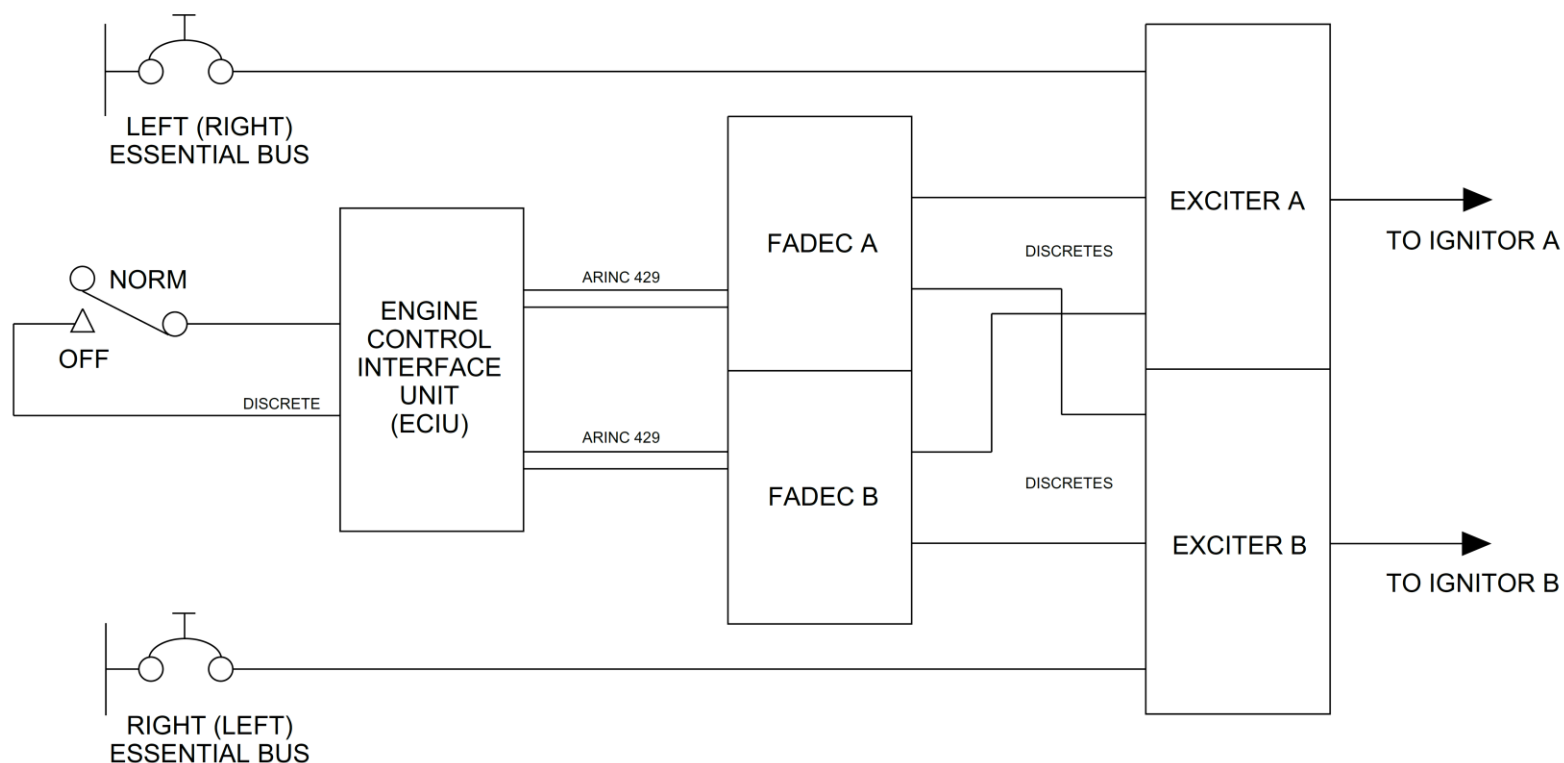


Figure - Ignition Power Supply Schematic

Ignition Distribution

Introduction

The ignition distribution system transmits the dc pulsed high voltage output from the ignition exciter to the ignition plugs and converts this voltage to sparks that are used to ignite the fuel/air mixture.

General Description

The function of the ignition cables is to transmit the high voltage signal from the ignition exciter to the ignition plugs. The function of the ignition plugs is to convert the high voltage signal to high temperature sparks to ignite the fuel/air mixture.

Detailed Description

When 28 Vdc is supplied to the ignition exciter the unit outputs a pulsed high voltage signal which is carried by the ignition cables to the ignition plugs. This

voltage is then transformed into sparks which ignite the fuel/air mixture.

Ignition Distribution – Component Description

Ignition Cables

[Refer Figure - Ignition Cables](#)

The cables are mounted on the right side of the engine. The cables are flexible and provide for a shielded electrical connection between the ignition exciter and the ignition plug. Each cable is constructed from a 5 mm silicon jacketed conductor wire enclosed by an inner braid of nickel-clad copper wire. This is further enclosed by an inner core braided with nickel wire. The end fittings are stainless steel and the inner core of the cable is hermetically sealed. This is to provide protection against electromagnetic interference.

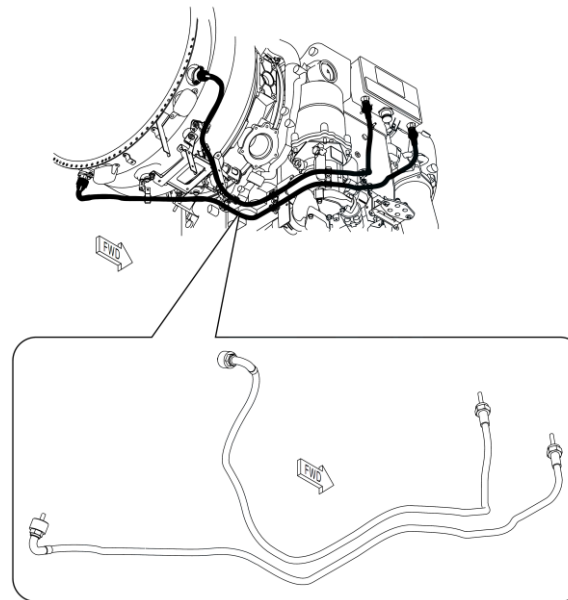


Figure - Ignition Cables

Ignition Plugs

Refer Figure - Ignitor Plug

There are two ignition plugs per engine. They are installed in the gas generator case at the 4 and 7 o'clock positions. The plugs are of the drop-in type and are secured to the engine by installing them between the gas generator case and the ignition cable fittings. This reduces the time required to replace the plugs. Each plug has a central electrode within a shell which acts as the ground.

As the energy potential increases, a small current passes across the semi-conducting material until the air between the electrode and the shell ionizes. At this point, high energy voltage discharges across the gap creating a spark. The spark always occurs between the electrode and the shell.

The plug shells are made from Hastelloy X for creep and fret resistance. The electrodes are made from Iridium for maximum life. Holes in the igniter body let cooling air pass to the igniter tip. This cooling air helps to keep the igniter tip cool.

Training Information Points

Remove the power to the ignition exciter and wait six (6) minutes before you touch the ignition cables. The voltage in the cables can injure or kill you if touch them with the ignition exciter powered on.

The ignition plugs are fragile. If they are dropped during handling they must be replaced with new ones. This is because damage to the interior of the plug cannot be seen from the outside.

The ignition cable fittings are coated at installation to ensure good electrical conductivity.

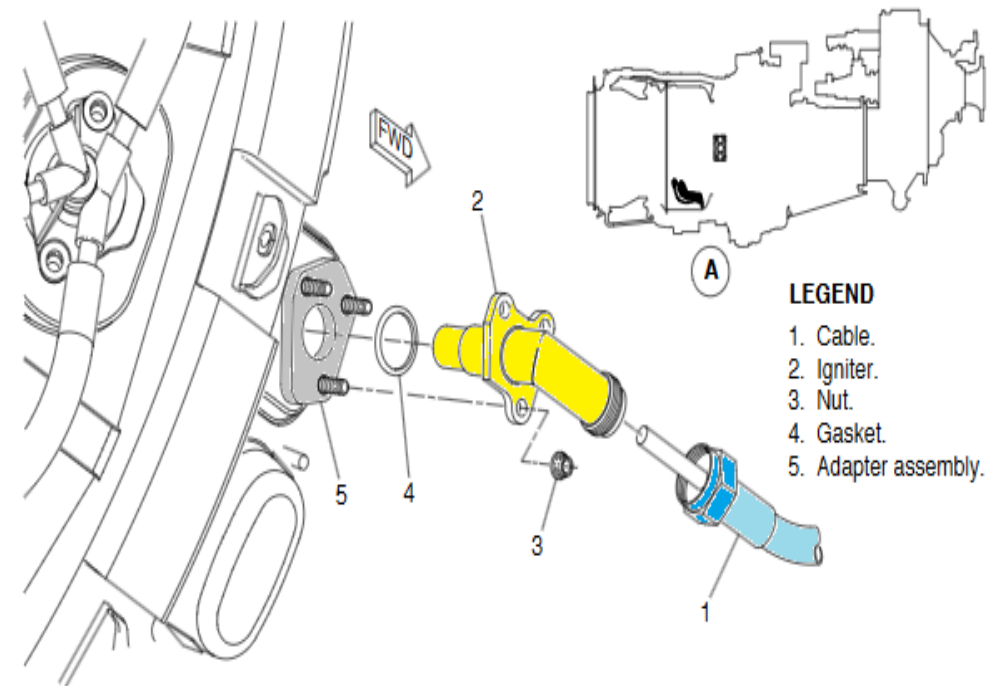


Figure - Ignitor Plug

Ignition Selection

Introduction

The ignition selection system selects the engine to be started and transmits this instruction to the Full Authority Digital Electronic Control (FADEC). The system is also used to set the ignition system mode to either “NORMAL” or “OFF”

General Description

The function of the ignition selection system is performed by the ignition selection switches. A panel mounted in the flight compartment has the switches that are used for engine ignition selection. This panel is labeled ENGINE START and has the switches that follow:

- Ignition Selection Switches
- Engine starter select switch
- Engine start push switch

Detailed Description

[Refer Figure - Engine Start Panel & Ignition Selector Switch](#)

Switches on the flight compartment instrument panel are used to select which engine is started and in what mode the ignition system is set to operate. The ignition selection system operates in two modes. They are the “OFF” mode and the “NORMAL” mode

Ignition Selection Switches

The ignition selection switches allow the flight crew to select the ignition mode. There are two ignition modes; “NORMAL” and “OFF”. When the switches are in the “NORMAL” position, the FADEC activates ignition during engine starts (on the ground or in-flight). When the switches are in the “OFF” position, the FADEC disables ignition regardless of the ground or flight status. The FADEC can command both igniters to operate during flameout or for surge accommodation.

Engine Starter Select Switch

The engine starter select switch is used to select the left or right engine

starter. The switch is a three position toggle switch with a left, center and right positions. The center position is neutral.

Engine Start Push switch

This switch signals the related FADEC to activate the selected engine’s starter.

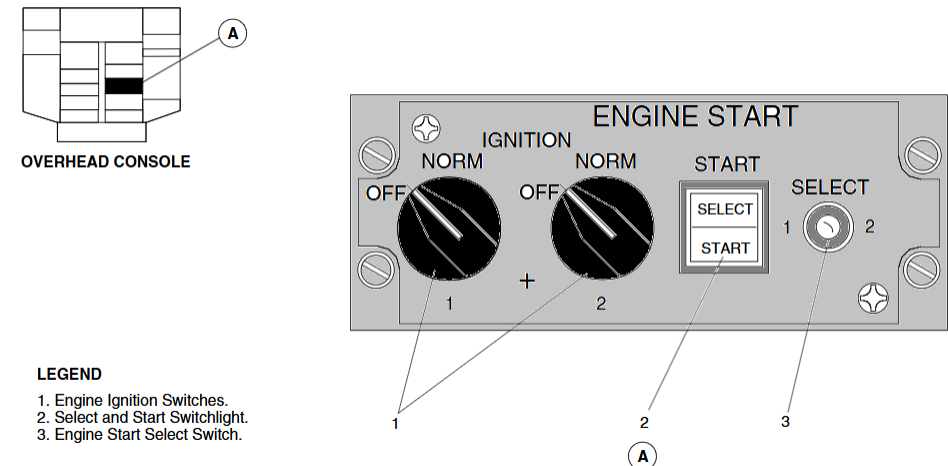


Figure - Engine Start Panel & Ignition Selector Switch

Ignition System Operation

[Refer Figure - Ignition System Operation](#)

Normal System Operation

Each exciter has a channel "A" and a channel "B". The left essential bus supplies electrical power to channel "B". The right essential bus supplies electrical power to channel "A". With the BATTERY MASTER selected on the exciters are powered and wait for a signal from the FADEC to operate.

The switch on the engine start control panel when in the OFF position provides continuity "OFF" discrete to channel A and B of the ECIU. In the Norm Position it provides a failsafe "ON" discrete to channel A and B of the ECIU. The ECIU changes the analog signal to a digital signal and each channel sends a signal to channel A and B of each FADEC on an ARINC 422 bus.

FADEC will default to normal if it loses communication with the ECIU.

When FADEC detects N_h is greater than 8%, it sends a discrete to the exciter and commands one ignitor to operate. Each channel of FADEC can operate either ignitor plug. On subsequent starts the other ignitor is used., this is to minimize ignitor wear. After the start, N_h greater than 50%, FADEC will select the ignitor "Off".

Abnormal Operation

FADEC will command both ignitors to operate if:

- FADEC detects no light off after 8 seconds from selected Fuel On
- FADEC detects there is a flameout when airborne
- FADEC detects that there is a surge or compressor stall

If an ignitor does not operate, FADEC will log a fault count of 3 to the ignitor. If the ignitor reaches a count of 9 then a fault code is logged for the faulty ignitor.

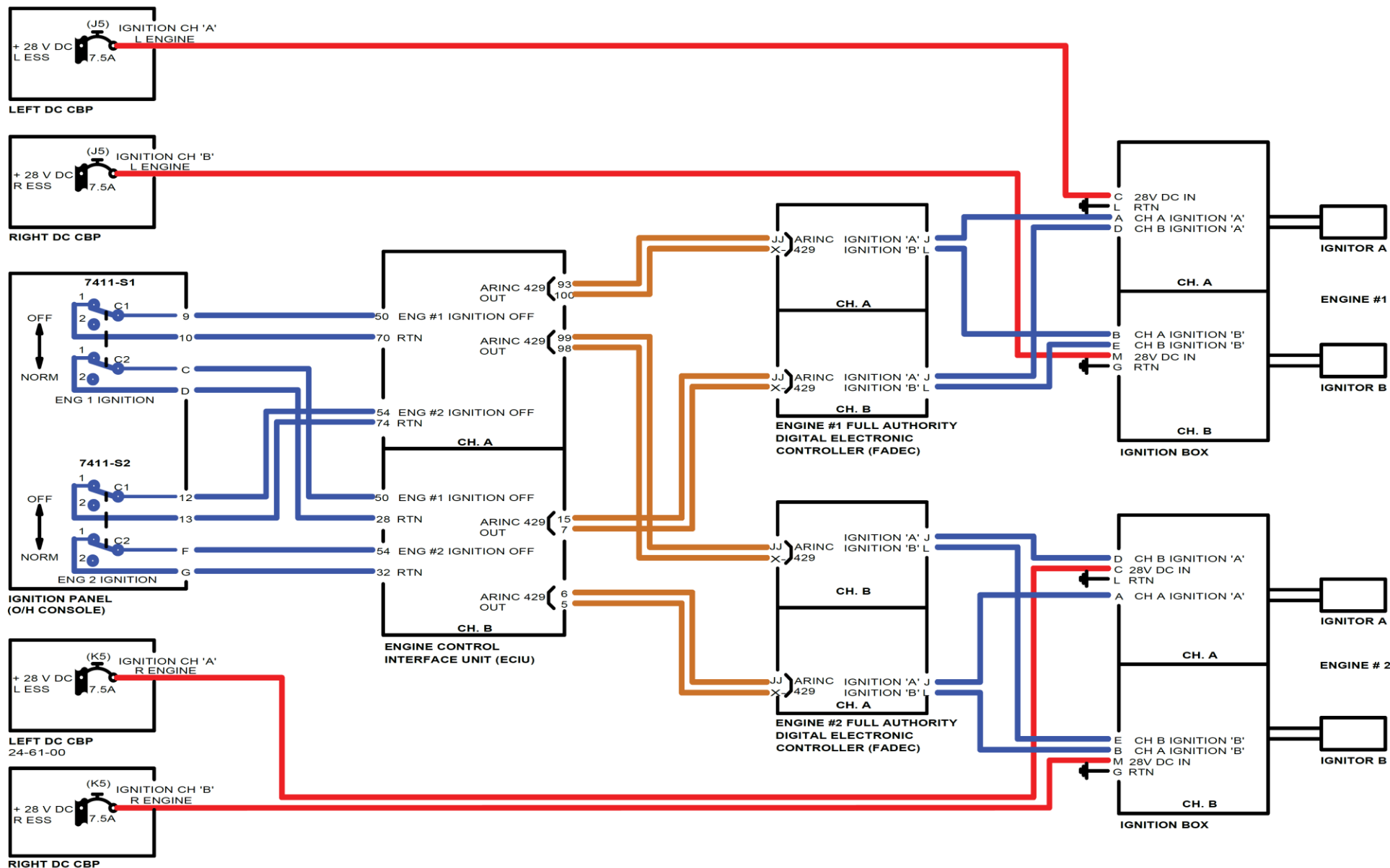


Figure - Ignition System Operation – Electrical Schematic